COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 11 : SLO 2 — IEEE 754 Double Precision Format
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Section A: Match the Following (Remember)

• A. Total bits in double precision → 3. 64 bits
• B. Sign bit → 4. Indicates positive or negative
• C. Exponent field → 1. 11 bits
• D. Mantissa (fraction) → 2. 52 bits
• E. Exponent bias in double precision → 5. 1023

Section B: Fill in the Blanks (Understand)

• Double precision format uses a total of 64 bits.
• The exponent field in double precision is 11 bits long.
• The exponent is stored with a bias of 1023 in double precision format.
• The mantissa is composed of 52 bits and represents the fractional part.
• The sign bit is set to 0 for positive numbers and 1 for negative numbers.

Section C: Activity — Encode -25.625 in IEEE 754 Double Precision

• Convert 25.625 to binary: 25 = 11001, 0.625 = 0.101 → 11001.101
• Normalize: 1.1001101 × 2^4
• Sign bit: 1 (negative number)
• Biased exponent: 4 + 1023 = 1027 = 10000000011
• Mantissa (52 bits): 1001101 0000000000000000000000000000000000000000000000000 (padded with zeros)
• Full 64-bit representation: 1 10000000011 1001101000000000000000000000000000000000000000000000